

Project: AttendanceSystem

This is a full-stack attendance monitoring system. The frontend is built with .NET Blazor WebAssembly, and the backend is a PHP API that connects to a database.

1. Software Requirements

To run this project, you will need the following software installed:

- **.NET 10 SDK (or newer):** Required to build and run the Blazor frontend application. You can download it from the official .NET website.
 - **A Web Server Stack (XAMPP or WAMP):** The backend is written in PHP and requires a web server and a database. XAMPP is a free and easy-to-install Apache distribution containing MariaDB (a community-developed fork of MySQL), PHP, and Perl.
 - **Apache:** To host the PHP API.
 - **PHP:** To execute the backend logic.
 - **MySQL/MariaDB:** To store the attendance data.
 - **Code Editor/IDE:**
 - **Visual Studio 2022 (or newer):** Recommended for a seamless .NET development experience.
 - **Visual Studio Code:** A great alternative with the **C# Dev Kit** extension installed.
-

2. Installation and Setup Steps

Follow these steps to get the project running on a new machine.

Step 1: Set Up the Backend (PHP API & Database)

1. **Install XAMPP:**
 - Download and install XAMPP from the Apache Friends website.
 - Run the **XAMPP Control Panel**.
2. **Start Services:**
 - Start the **Apache** and **MySQL** services from the XAMPP Control Panel.
3. **Deploy the PHP API:**
 - Navigate to your XAMPP installation directory (e.g., C:\xampp).

- Find the htdocs folder (e.g., C:\xampp\htdocs). This is the web server's root directory.
- Copy your entire attendance-api folder into the htdocs directory.
- You should now be able to access your API scripts in a browser, for example: http://localhost/attendance-api/get_logs.php.

4. Create the Database:

- Open your web browser and navigate to <http://localhost/phpmyadmin>.
- Create a new database (e.g., attendance_db).
- Import your database schema (the .sql file) to create the necessary tables. If you don't have a .sql file, you will need to create the tables manually based on your project's needs.

5. Configure the Database Connection:

- Open your PHP API files (e.g., [get_logs.php](#) and any database connection file like db_connect.php).
- Update the database connection settings (hostname, username, password, database name) to match your local XAMPP setup.

Step 2: Set Up the Frontend (Blazor Application)

1. Install .NET SDK:

- If not already installed, download and install the **.NET 10 SDK** from the official Microsoft website.

2. Open the Project:

- Open the AttendanceSystem.sln file in Visual Studio, or open the project folder in VS Code.

3. Restore Dependencies:

- The IDE should automatically restore the required NuGet packages. If not, you can do it manually by running the following command in the project's root directory:

```
bash
```

```
dotnet restore
```

4. Run the Application:

- You can now run the project. In Visual Studio, press the "Run" button (or F5). In VS Code or the terminal, run:

```
bash
```

`dotnet watch run`

- Your web browser will open, and the Blazor application will be running. It will make requests to the PHP backend you set up at <http://localhost/attendance-api/>.